UNIT- 02

GRAPHIC CARDS & I/O DEVICES

GRAPHICS CARDS

A **graphics card (GPU – Graphics Processing Unit)** is a hardware component that processes images, videos, and animations and sends them to the monitor.

Types of Graphics Cards

1. Integrated Graphics Card

An **integrated graphics card** is built into the motherboard or CPU.

Features:

1. Built inside processor (like Intel UHD Graphics((Ultra High Definition))
2. No separate hardware required
3. Uses system RAM
4. Low power consumption
5. Low performance compared to dedicated GPU

Examples:

•Intel UHD Graphics

Best for:- Office work, Browsing, Video watching, Basic programming

2. Dedicated Graphics Card

A **dedicated graphics card** is a separate hardware component installed in the motherboard.

Features:

1. Separate GPU
2. Has its own VRAM (Video RAM)
3. High performance
4. Required for gaming, AI, video editing

Examples:

1. NVIDIA GeForce - **NVIDIA GeForce** is a dedicated GPU series that provides high-performance graphics processing for gaming, video editing, and 3D applications.
2. AMD Radeon – (Advanced Micro Devices Radeon) **AMD Radeon** is a GPU product line that delivers graphics performance for gaming, multimedia, and content creation.

Best for:- Gaming, Video editing, 3D modelling, Cyber security GPU cracking tasks, Machine learning

Type of GPUs (Graphics Processing Unit)

Dedicated GPU



Dedicated GPUs use a separate PCB and use dedicated VRAM

Integrated GPU (iGPU)



iGPUs are integrated with the CPU and use system memory (RAM)

Installation of Graphics Card

Steps:

Step 1: Turn off computer

1. Shutdown system
2. Remove power cable

Step 2: Open CPU cabinet

1. Remove side panel using screwdriver

Step 3: Locate PCIe Slot

1. Find **PCI Express x16 slot** on motherboard

Step 4: Insert graphics card

1. Carefully align GPU with slot
2. Press gently until locked

Step 5: Connect power cable

1. (Some GPUs require 6-pin or 8-pin power)

Step 6: Tighten screw

1. Secure GPU to cabinet

Step 7: Close cabinet

Step 8: Install drivers

1. Install:
2. NVIDIA drivers
3. AMD drivers

Expansion Cards

Expansion cards are hardware cards inserted into motherboard slots to add new features.

Common Types of Expansion Cards

Expansion Card	Function
Graphics Card	Video processing
Sound Card	Audio processing
Network Card	Internet connection
WiFi Card	Wireless connectivity
Capture Card	Video recording
Storage Controller	Extra storage ports

External Devices (Peripheral Devices)

External devices are hardware connected outside the computer.

- **Types of External Devices**
- **Input Devices**- Used to give input.
- Examples:- Keyboard, Mouse, Scanner, Microphone, Webcam

- **Output Devices**- Used to display output.
- Examples:- Monitor, Printer, Speaker, Projector

- **Storage Devices**- Used for data storage.
- Examples:- Pen drive, External HDD, Memory card

Difference Between Expansion Cards and External Devices

Expansion Cards	External Devices
Installed inside CPU	Connected outside
Uses motherboard slots	Uses USB or ports
Permanent installation	Easy to remove
Example: GPU	Example: Keyboard

Monitor

A **monitor** is an output device that displays images, videos, and text.

Types of Monitors

1. LCD Monitor- Liquid Crystal Display
A flat panel display technology that uses liquid crystals to control light and display images.
Uses Liquid Crystal Display
Thin and lightweight
Low power consumption

2. LED Monitor- Light Emitting Diode
A display technology that uses LED backlighting to improve brightness and reduce power consumption.
Improved version of LCD
Better brightness
Better color quality

3. OLED Monitor- Organic Light Emitting Diode
An advanced display technology where organic materials emit light directly for better contrast and color.
High quality display
Expensive
Used in professional work

Monitor

A **monitor** is an output device that displays images, videos, and text.

Monitor Ports

Common ports:- HDMI, VGA(Video Graphics Array), DisplayPort, USB-C(Universal Serial Bus Type-C)

DP – DisplayPort

A digital display interface mainly used for connecting high-resolution monitors.

HDMI – High Definition Multimedia Interface

A digital interface used to transmit high-quality audio and video signals between devices.

VGA – Video Graphics Array

An analog display interface used to connect monitors to computers (older technology).

USB-C – Universal Serial Bus Type-C

A modern reversible connector used for data transfer, video output, and power delivery.

Keyboard

A **keyboard** is an input device used to enter data.

Types of Keyboards

1. Membrane Keyboard

- Cheap
- Quiet
- Used in offices

2. Mechanical Keyboard

- Uses switches
- Durable
- Used by programmers and gamers

3. Wireless Keyboard

- Uses Bluetooth or USB receiver

Keyboard Sections

- Alphanumeric keys
- Function keys
- Numeric keypad
- Control keys

Mouse

A **mouse** is a pointing input device.

Types of Mouse

1. Wired Mouse
•Uses USB cable
•Reliable

2. Wireless Mouse
•Uses battery
•Uses Bluetooth or dongle

3. Optical Mouse
•Uses LED sensor

4. Laser Mouse
•High precision

DPI – Dots Per Inch- A measurement of mouse sensitivity indicating how far the cursor moves relative to physical movement.
LED Sensor – Light Emitting Diode Sensor A sensor used in optical mice to detect movement using light reflection.
Bluetooth- A wireless communication technology used to connect devices over short distances.

Power Supplies and Cooling Systems

Power Supply Unit (PSU)

A **PSU** converts AC power into DC power for computer components.

Functions of PSU

- Provides power to motherboard
- Powers CPU
- Powers GPU
- Powers storage devices

PSU Types

1. Non-Modular PSU
All cables fixed.

2. Semi-Modular PSU
Some removable cables.

3. Fully Modular PSU
All cables removable.

Selecting and Installing PSU

How to Select PSU

1. Wattage
Typical requirements:

PC Type	PSU Required
Basic PC	450W
Gaming PC	650W
High end PC	850W+

Selecting and Installing PSU

How to Select PSU

1. Wattage

Typical requirements:

PC Type	PSU Required
Basic PC-	450W
Gaming PC-	650W
High end PC-	850W+

2. Efficiency Rating

Look for:

•80+ Bronze
•80+ Silver
•80+ Gold

3. Connectors Required

Check:

•24-pin motherboard
•8-pin CPU
•GPU connectors

PSU Installation Steps

Step 1:- Open cabinet
Step 2:- Place PSU in PSU bay (Power Supply Unit)
Step 3:- Align screw holes
Step 4:- Tighten screws
Step 5:- Connect cables:

1. *Motherboard power*
2. *CPU power*
3. *GPU power*
4. *SATA power*

COOLING PRINCIPLES AND COOLING COMPONENTS

COMPUTERS PRODUCE HEAT. COOLING PREVENTS DAMAGE.

Cooling Principles

Cooling works using:

1. Air Cooling

Uses fans.

2. Liquid Cooling

Uses coolant.

3. Thermal Conductivity

Heat moves from CPU to heatsink

Cooling Components

1. **Heat Sink-** Absorbs heat.
2. **Cooling Fan-** Moves air.
3. **Thermal Paste-** Improves heat transfer.
4. **Liquid Cooler-** Uses radiator and pump.

Cooling Related Terms

- **RPM – Revolutions Per Minute**
  A unit measuring the speed of rotating cooling fans.
- **Thermal Paste (TIM – Thermal Interface Material)**
  A heat-conductive compound placed between CPU and heatsink to improve heat transfer.
- **Heat Sink**
  A metal component that absorbs and dissipates heat from processors.
- **Cooling Fan**
  A device that moves air to reduce component temperature.
- **Liquid Cooling System**
  A cooling method that uses coolant circulation to remove heat.

CABLE MANAGEMENT AND AESTHETICS

CABLE MANAGEMENT MEANS ORGANIZING CABLES PROPERLY INSIDE CABINET..

Benefits:

1. Better airflow
2. Better cooling
3. Easy maintenance
4. Professional look

Cable Management Methods:

- Use cable ties
Bundle cables.
- Route behind motherboard
Hide cables.
- Use modular PSU
Reduces extra cables.
- Use cable channels
Organize wiring.

Cable Management Terms

- **Modular PSU**
  A PSU where cables can be detached to reduce clutter.
- **Cable Tie (Zip Tie)**
  A plastic fastener used to organize cables.
- **Airflow**
  Movement of air inside the case to maintain cooling.
- **Aesthetics**
  The visual appearance and neatness of the PC build.


BEFORE
AFTER


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CX600M

ASSEMBLING A PC

PC ASSEMBLY MEANS CONNECTING ALL HARDWARE COMPONENTS.

Components Required:

1. Cabinet
2. Motherboard
3. CPU
4. RAM
5. Storage
6. PSU
7. GPU
8. Cooling fans

Step-by-Step Guide to Assembling a PC

Step 1: Install CPU- Place CPU carefully into socket.
Step 2: Apply thermal paste- Small drop on CPU.
Step 3: Install CPU cooler- Fix heatsink and fan.
Step 4: Install RAM- Insert into RAM slots.
Step 5: Install motherboard- Place inside cabinet.
Step 6: Install PSU- Fix PSU.
Step 7: Install storage- Install SSD/HDD.
Step 8: Install GPU- Insert into PCIe slot.
Step 9: Connect cables- Connect all power cables.
Step 10: Power ON test- Check POST screen.

PC Assembly Related Terms

- **POST – Power On Self Test**
  A diagnostic process performed when a computer starts to check hardware functionality.
- **BIOS – Basic Input Output System**
  Firmware that initializes hardware during booting.
- **UEFI – Unified Extensible Firmware Interface**
  Modern replacement for BIOS with improved features and graphical interface.
- **PCIe – Peripheral Component Interconnect Express**
  A high-speed interface standard used for GPUs and expansion cards.
- **SATA – Serial Advanced Technology Attachment**
  An interface used to connect storage devices like HDD and SSD.
- **M.2 Slot**
  A motherboard slot used for high-speed SSD installation.
- **Motherboard**
  The main circuit board that connects all computer components.
- **Cabinet (Computer Case)**
  The enclosure that houses computer components.

Precautions Tips

Precautions are safety measures taken to prevent damage to computer components.

Important precautions:

- **1. Disconnect power supply**
  Always switch off and unplug the computer before working inside it.
- **2. Use anti-static protection**
  Use an anti-static wrist strap to prevent electrostatic discharge (ESD) damage.
- **3. Handle components carefully**
  Always hold components by edges and avoid touching circuits.
- **4. Check compatibility**
  Ensure motherboard, RAM, CPU, and PSU are compatible.
- **5. Use proper tools**
  Use correct screwdrivers and avoid forcing components.
- **6. Maintain clean workspace**
  Work in a dust-free and dry environment.

Troubleshooting Tips

Troubleshooting means identifying and solving hardware or software problems.

- **Common troubleshooting situations:**

Computer not starting

Check power cable

Check PSU switch

Check motherboard connections

No display problem

Check monitor cable

Check GPU installation

Check RAM seating

Overheating
Check cooling fan
Apply thermal paste
Improve airflow

OS not installing
Check boot order
Check installation media
Check storage connection

System beeping sound
RAM problem
GPU issue
Motherboard error codes

Hardware Troubleshooting & Diagnosing Issues

- **What is Hardware Troubleshooting?**
  Hardware troubleshooting is the process of identifying, isolating, and fixing problems related to physical components of a computer.
- **What is Diagnosing?**
  Diagnosing means finding the exact cause of a hardware problem before fixing it.

Hardware Troubleshooting & Diagnosing Issues

- **Troubleshooting Process**

Step 1: Identify the Problem

1. What is not working?
2. When did the issue start?

Observe symptoms: No power, No display, Strange noise, Slow performance

Step 2: Check Basic Things First

Power cable connected?, Switch ON?, Monitor working?, Loose connections?

Step 3: Isolate the Problem

1. Remove unnecessary components
2. Test one component at a time
3. Example: Remove GPU → check display, Change RAM slot

Step 4: Diagnose Using Clues

- **Beep Codes (from motherboard):**
  Continuous beep - RAM issue , No beep - motherboard/CPU issue
- **Error Messages:** “No boot device” - storage problem
- **Visual Signs:** Burning smell - PSU issue, Dust – overheating

Step 5: Fix the Problem

Replace faulty component, Reconnect cables, Clean system, Install/update drivers

Step 6: Verify Solution

Restart system, Check if issue is solved, Monitor performance

Common Problems & Solutions

1. Computer Not Turning ON

Causes: PSU failure, Loose power cable, Faulty motherboard

Solution: Check power cable, Check PSU switch, Try another PSU

2. No Display Problem

Causes: Loose RAM, GPU issue, Monitor cable issue

Solution: Reseat RAM, Check GPU, Replace cable

3. Continuous Beeping Sound

Causes: RAM not installed properly, Hardware failure

Solution: Reinsert RAM, Replace faulty RAM

4. Overheating Problem

Causes: Dust, Fan not working, No thermal paste

Solution: Clean system, Check fan, Apply thermal paste

5. Slow Performance

Causes: Low RAM, Background apps, Malware

Solution: Upgrade RAM, Close apps, Scan system

Replacing Faulty Components-

Replacing Faulty Components is the process of removing damaged or non-working hardware parts and installing new ones to restore proper functioning of a computer.

1. Turn OFF and unplug the system
Prevents electric shock and damage
2. Open the cabinet (CPU case)
Use screwdriver carefully
3. Identify faulty component
Based on troubleshooting
4. Remove the component
Gently disconnect cables
Avoid force

5. Install new component
Align properly
Insert firmly
6. Reconnect cables
Ensure proper connections
7. Close cabinet and power ON
Test the system
Check if problem is solved

Common Components

1. RAM (Random Access Memory)

Signs of Faulty RAM: System not starting, Continuous beep sound, Frequent crashes

Replacement Steps: Open RAM clips, Remove old RAM, Insert new RAM properly (click sound)

2. HDD / SSD (Storage Device)

Signs of Faulty Storage: OS not loading, Slow system, Disk errors

Replacement Steps: Remove SATA cable and power cable, Unscrew drive, Install new HDD/SSD, Reconnect cables

3. PSU (Power Supply Unit)

Signs of Faulty PSU: No power, Random shutdown, Burning smell

Replacement Steps: Disconnect all cables, Unscrew PSU, Install new PSU, Reconnect cables carefully

4. GPU (Graphics Card)

Signs of Faulty GPU: No display, Screen flickering, Low performance

Replacement Steps: Remove screw and PCIe lock, Pull out GPU, Insert new GPU, Connect power cable

5. Cooling Fan

Signs of Fault: Overheating , Fan not spinning

Replacement Steps: Disconnect fan cable, Unscrew fan, Install new fan

THE END

THANKYOU